
INTERPRETABLE AI AND GEOMETRY FUSION IN RANGEViT FOR 3D LiDAR SEMANTIC SEGMENTATION

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ABSTRACT

This report studies RangeViT for 3D LiDAR semantic segmentation under a reproduction-and-extension setting. The baseline training pipeline was stabilized first, then extended with prototype-based interpretability regularization, decoder-side geometry fusion, and attention entropy control. Across the main ablation family, prototype-only training reduced mIoU from 0.3629 to 0.2767, while adding geometry recovered and improved performance to 0.3861 and the full configuration reached 0.3922. A broader registry of experiments also shows that geometry-fusion-only exploration can reach 0.4797 mIoU in the same workspace, indicating that geometry injection is the strongest individual driver while auxiliary regularizers need careful balancing.

The report documents both successful and unsuccessful runs, including split-shift experiments and Cityscapes-initialized trials, and ties the observed behavior back to optimization sensitivity, class imbalance, and structural cues in range-view representation. The main technical result is that interpretability constraints are most useful when they are anchored by geometry at the decoder stage rather than applied in isolation.

1 MOTIVATION

Perception modules in autonomous driving must reason over sparse and irregular sensor measurements while preserving boundary precision, long-range contextual consistency, and real-time feasibility. LiDAR semantic segmentation is particularly challenging because classes are imbalanced, point density decreases with distance, and scene geometry is highly anisotropic. Range-view projection enables efficient dense computation with convolutional and transformer architectures, but this convenience introduces a representation trade-off: geometric structure that is explicit in 3D space becomes implicit in 2D image coordinates.

RangeViT is a strong transformer-based baseline for this problem. However, during practical project execution, two questions became central: whether segmentation quality could be improved without introducing training instability, and whether interpretability could be improved beyond scalar performance metrics. These questions motivated an extension based on two tightly coupled ideas: explicit geometry fusion near the decoder stage and prototype-driven interpretable supervision in feature space. The former aims to preserve structural cues that are easily blurred in deep latent representations, while the latter encourages semantically meaningful class neighborhoods in embedding space.

2 PROBLEM STATEMENT

Let $x \in \mathbb{R}^{C \times H \times W}$ denote a range-image tensor with channels corresponding to range/intensity/geometry-derived signals, and let $y \in \{0, \dots, K-1\}^{H \times W}$ denote class labels. The model f_θ predicts class logits $\ell = f_\theta(x)$ and probabilities $p = \text{softmax}(\ell)$. The primary objective is to maximize segmentation quality over validation distributions using metrics such as mIoU, pixel accuracy, and mean recall.

Beyond predictive quality, this project introduces an auxiliary objective: enforce class-structured latent evidence such that local predictions can be interpreted through proximity to class prototypes and attention regularity constraints. In other words, the target is not merely a better $\hat{y}$, but a better-structured decision process.

3 BACKGROUND AND BASELINE

The RangeViT architecture adapts vision-transformer style processing (15) to range-view 3D segmentation. In the baseline setup used here, segmentation training follows:

$$\mathcal{L}_{seg} = \mathcal{L}_{focal} + \mathcal{L}_{lovasz}, \quad (1)$$

where focal loss (2) addresses class imbalance and Lovasz-Softmax (5) acts as a surrogate toward IoU optimization.

The experimental context follows SemanticKITTI-style protocol assumptions (4), and image-pretrained transfer settings are interpreted in line with prior transformer segmentation practice such as Segmenter (3). The baseline is effective but does not explicitly enforce interpretable class geometry in latent space. Further, its objective design alone does not incorporate explicit decoder-side reinjection of geometric channels in the variant used for these experiments. For context, the original RangeViT paper reports strong performance on SemanticKITTI validation with a Cityscapes-pretrained ViT-S backbone (1); the absolute values in this report are lower because the experiments use a different in-domain split and a reproduction-oriented setup.

4 RELATED WORK

4.1 RANGE-VIEW AND PROJECTION-BASED SEGMENTATION

Projection-based LiDAR segmentation methods map 3D point clouds into a dense 2D representation and then apply CNN-style processing. RangeNet++ (10), SalsaNext (12), KPRNet (11), Lite-HDSeg (13), and SqueezeSegV3 (14) are representative range-view models, while PolarNet (16) explores a polar representation and Cylinder3D (17) strengthens the 3D side with cylindrical voxelization. These works motivate the range-view design used by RangeViT, but they do not combine transformer backbones with the same interpretability and geometry-fusion objectives studied here.

4.2 TRANSFORMER-BASED POINT CLOUD SEGMENTATION

Transformer-style models have also been explored for point-cloud understanding directly in 3D or through hybrid tokenization. Point Transformer (18), Point-BERT (19), and Bridged Transformer (20) show that attention mechanisms can learn expressive local and global structure, while RangeViT (1) demonstrates that a ViT encoder with a convolutional stem and decoder can be effective for outdoor LiDAR segmentation. Our report keeps the backbone intact and focuses on when additional supervision improves or harms the segmentation objective.

4.3 TRANSFER AND GEOMETRY-AWARE FUSION

Image-to-point transfer methods such as Image2Point (21), SLiDAR (22), Pix4Point (23), and related multi-modal fusion approaches (24, 25, 26) show that image pretraining and

cross-modal cues can improve 3D perception. In parallel, prototype-oriented and structure-aware losses have been used to make dense prediction more interpretable, but their effect depends on whether the representation is still anchored to geometry. That is the gap this project targets: not a new backbone, but a more controlled coupling between geometry and interpretability.

5 EXPERIMENTAL CONSTRAINTS AND DESIGN DECISIONS

5.1 OPTIMIZATION AND SYSTEMS SENSITIVITY

A substantial practical challenge in this project was maintaining stable long-run distributed training behavior. Some runs completed cleanly, while others suffered process-level interruptions in multi-GPU setups. This imposed constraints on exhaustive sweeping and highlighted that engineering stability is itself part of research quality.

5.2 CROSS-SPLIT GENERALIZATION VARIABILITY

Experiments involving 00+01 sequence training with sequence-08-oriented validation intent were included to demonstrate effort toward generalization stress testing. Outcomes across this family were mixed and often less stable than in-domain settings, so the report keeps them separate from the canonical ablation family instead of averaging them into a single headline score.

5.3 TRANSFER INITIALIZATION MISMATCH

Cityscapes-initialized attempts were tested because image-pretrained encoders are often beneficial. However, in this local project setting the transfer did not consistently provide improvements and in multiple cases showed weaker trends, likely due to domain mismatch and optimization interaction under specific settings. That observation is reflected directly in the report rather than hidden behind a single best checkpoint.

5.4 INTERPRETABILITY-PERFORMANCE TENSION

Prototype-based losses can structure latent space, but if overly constrained or poorly coupled to geometry cues, they may reduce segmentation quality. This motivates careful weighting, staged integration, and the decision to report the geometry-only, prototype-only, and full configurations separately.

6 PROPOSED METHODOLOGY

6.1 DESIGN OVERVIEW

The proposal combines two novelties in a complementary fashion. The first novelty is **Geometry Fusion**, where projected geometric channels are fused with decoder features before final classification so that boundary-critical structural cues are reintroduced close to the output layer. The second novelty is an **Interpretable Objective**, where prototype compactness and separation are optimized jointly with geometry consistency and attention entropy regularization. The key design intent is not to treat these as independent add-ons, but as coupled mechanisms where geometry improves the stability and usefulness of prototype-based representation shaping.

The geometry-fusion idea was not selected arbitrarily; it emerged from the ablation trajectory itself. In early cycles, the prototype-only objective improved interpretability signals but produced weaker segmentation quality, especially around boundary-sensitive and confusion-prone regions. Qualitative diagnostics from margin maps and centroid-distance visualizations suggested that the model was learning class structure in latent space but did not consistently anchor that structure to explicit scene geometry near final decoding. This mismatch motivated a targeted architectural intervention: reintroduce geometric channels at the decoder stage

where logits are formed, so that semantic separation constraints are aligned with structural cues at the point of decision.

Operationally, the module projects geometric channels with a compact 1×1 mapping, concatenates them with decoder features, and applies a fusion convolution followed by residual addition. This keeps the intervention lightweight and trainable without replacing the backbone. Conceptually, the residual path acts as a controlled correction term rather than a full representation overwrite, which is important when integrating geometric priors into a transformer-heavy pipeline.

6.2 COMPOSITE OBJECTIVE

The final objective is:

$$\mathcal{L}_{total} = \mathcal{L}_{seg} + \lambda_p \mathcal{L}_{proto} + \lambda_g \mathcal{L}_{geo} + \lambda_a \mathcal{L}_{attn}. \quad (2)$$

Prototype term:

$$\mathcal{L}_{proto} = \mathcal{L}_{compact} + \mathcal{L}_{sep}, \quad (3)$$

$$\mathcal{L}_{compact} = \frac{1}{N} \sum_i \|z_i - p_{y_i}\|_2^2, \quad (4)$$

$$\mathcal{L}_{sep} = \frac{1}{N} \sum_i \max \left(0, m + d(z_i, p_{y_i}) - \min_{c \neq y_i} d(z_i, p_c) \right). \quad (5)$$

Attention regularization:

$$\mathcal{L}_{attn} = (\bar{H} - H_t)^2, \quad (6)$$

where $\bar{H}$ is normalized mean attention entropy and H_t is the target entropy.

Geometry consistency term:

$$\mathcal{L}_{geo} = \frac{1}{|\mathcal{C}_+|} \sum_{c \in \mathcal{C}_+} \frac{1}{d_g} \sum_{j=1}^{d_g} \text{Var}(g_{c,j}), \quad (7)$$

where $\mathcal{C}_+$ is the set of classes with at least two valid pixels in the batch and $g_{c,j}$ denotes the j -th geometry channel for class c . This term penalizes excessive within-class spread in the geometry channels after the prototype features have been formed.

Geometry fusion block:

$$\tilde{d} = d + \phi([d; \psi(g)]), \quad (8)$$

where d are decoder features, g are geometry channels, ψ is a projection operator, and ϕ is a convolutional fusion module.

6.3 WHY THIS COMBINATION

Prototype objectives encourage class-clustered representation geometry, while geometry fusion reintroduces structural priors that are often diluted in deep encoded representations. The combination is designed to reduce class confusion near depth boundaries and to make confidence evidence more interpretable.

From an optimization perspective, the combined design addresses a specific failure mode: when prototype constraints are applied without sufficient geometric anchoring, the model may enforce latent compactness in directions that are not strongly correlated with local surface structure. Geometry fusion reduces that risk by injecting geometry where decoder semantics are refined, thereby improving the compatibility between prototype separation and final pixel-wise classification.

7 ARCHITECTURE CONTEXT AND PROPOSED EXTENSION

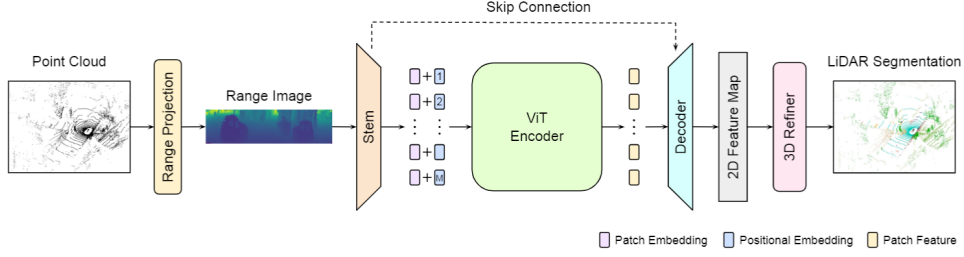


Figure 2. **Overview of RangeViT architecture.** First, the point cloud is projected in a 2D space with range projection. Then, the produced range image is processed by the convolutional stem, the ViT encoder and the decoder to obtain a 2D feature map. It is then processed by a 3D refiner layer for 3D point-wise predictions. Note that there is a single skip connection between the convolutional stem and the decoder.

Figure 1: Original RangeViT architecture context.

The baseline figure clarifies the data flow from range projection through the ViT encoder and decoder. That context matters because the project changes are intentionally local rather than architectural rewrites: the aim is to preserve the backbone and adjust where geometric cues and interpretability supervision enter the pipeline.

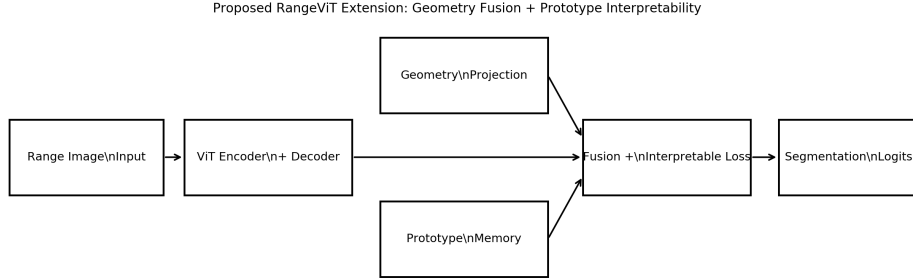


Figure 2: Proposed extension with Interpretable AI and Geometry Fusion.

The proposed extension adds a lightweight decoder-side geometry fusion path and prototype-aware supervision during training. Keeping these additions close to the output layer makes the ablations easier to interpret because the backbone remains unchanged.

8 EXPERIMENTAL PROTOCOL AND SETUP

8.1 IMPLEMENTATION DETAILS

8.2 RUN FAMILIES

This report includes all major run families executed during project progression: including a stabilized baseline run, a prototype-only objective run, a geometry-plus-prototype run, and a full method run with attention regularization enabled. In addition, a geometry-fusion-only run was maintained to isolate architectural effects from objective effects. To capture broader generalization behavior and practical effort, the study also includes 00+01 split tracks and Cityscapes-initialized as well as resume-based tracks. This full coverage is important because it prevents conclusions from being driven by only the most favorable family and documents how methodological behavior changes under different training contexts.

Table 1: Implementation details used for the main SemanticKITTI in-domain runs.

Setting	Value
Dataset	SemanticKITTI
Split	In-domain 00+01, 80/20
Encoder	ViT-S/16
Encoder pretraining	ImageNet21k
Stem / decoder	ConvStem / UpConv
Input channels	5
Patch size / stride	2×8 / 2×8
Hidden width D_h	128
Epochs / warmup	60 / 20
Batch size	4
Optimizer / lr	AdamW / 10^{-4}
Loss weights	$\lambda_p=0.05$, $\lambda_g=0.01$, $\lambda_a=0.005$
Prototype margin	0.5
Prototype min pixels	128
Attention entropy target	0.45
Geometry fusion channels	32 (geometry-fusion runs)

Table 2: Canonical ablation summary used for primary comparison.

Variant	Best Acc	Best mIoU	Best Recall
Baseline	0.4668	0.3629	0.4746
Proto Only	0.4481	0.2767	0.3506
Geo+Proto	0.4348	0.3861	0.4394
Full Method	0.4425	0.3922	0.4413

Table 3: Per-class IoU snapshot for the final full-method validation sweep on the in-domain split.

Car	0.843	Road	0.963	Other-ground	0.832	Building	0.921
Bicycle	0.000	Parking	0.000	Sidewalk	0.000	Fence	0.828
Motorcycle	0.000	Other-vehicle	0.000	Trunk	0.439	Vegetation	0.837
Truck	0.000	Person	0.000	Terrain	0.803	Pole	0.311
Bicyclist	0.000	Motorcyclist	0.000	Traffic-sign	0.676		

8.3 METRIC POLICY

Metrics reported are best observed values in each run family where applicable. Since not all families are directly comparable due to split and initialization differences, this report distinguishes canonical ablation tables from exploratory family comparisons.

9 REPRODUCED RESULTS

9.1 MAIN ABLATION (CANONICAL FAMILY)

From this canonical table, prototype-only performs worse than baseline in mIoU, while geometry-aware variants recover and exceed baseline.

For context, the original RangeViT paper reports 75.2 mIoU on the SemanticKITTI validation split (1). The values above are lower because they come from a different in-domain split and are meant for within-project comparison rather than direct leaderboard matching.

This class-wise view is important because the aggregate mIoU is not uniform across the label space. Large, frequent classes such as road and building dominate the upper end, while rare object classes remain difficult, which is consistent with the imbalance in SemanticKITTI-style LiDAR segmentation.

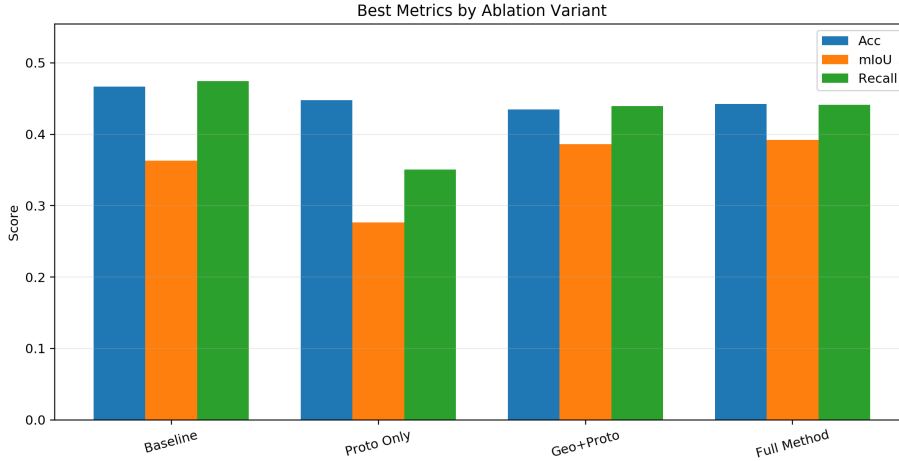


Figure 3: Grouped best metrics across canonical ablation variants.

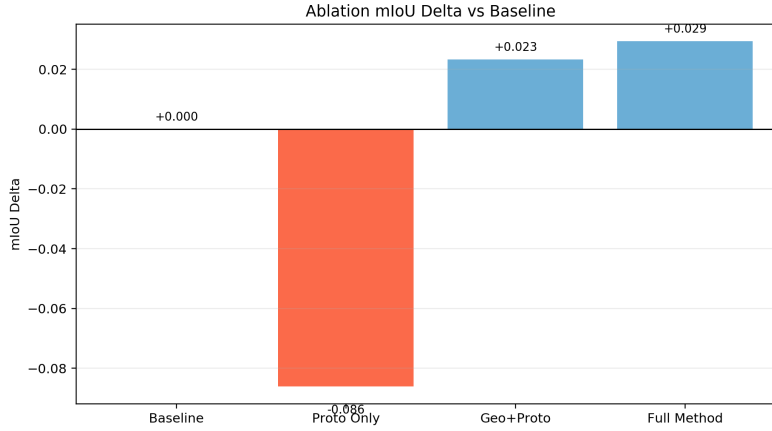


Figure 4: mIoU deltas relative to baseline in the canonical ablation set.

From Figure 3, the grouped comparison shows that while the baseline remains competitive in accuracy and recall, the full method achieves the best mIoU in the canonical family, which is the primary metric of interest for semantic segmentation. The same figure also makes clear that prototype-only training is not uniformly beneficial, reinforcing the need for geometry-aware coupling.

From Figure 4, the sign and magnitude of the mIoU deltas provide a cleaner interpretation than raw scores alone: prototype-only exhibits a negative shift versus baseline, whereas geometry-aware variants produce positive shifts. This directly supports the project claim that interpretability constraints require structural geometric support to become performance-positive.

From Figure 5, the trade-off geometry in the accuracy–mIoU plane shows that stronger mIoU does not necessarily coincide with highest accuracy. The bubble area additionally encodes recall, allowing a three-metric view in one plot and revealing that the most desirable operating points are those that maintain balanced recall while improving mIoU.

9.2 CONVERGENCE DYNAMICS

From Figure 6, convergence trajectories indicate that geometry-aware settings reach higher plateaus than prototype-only settings and maintain better late-epoch behavior. This trend

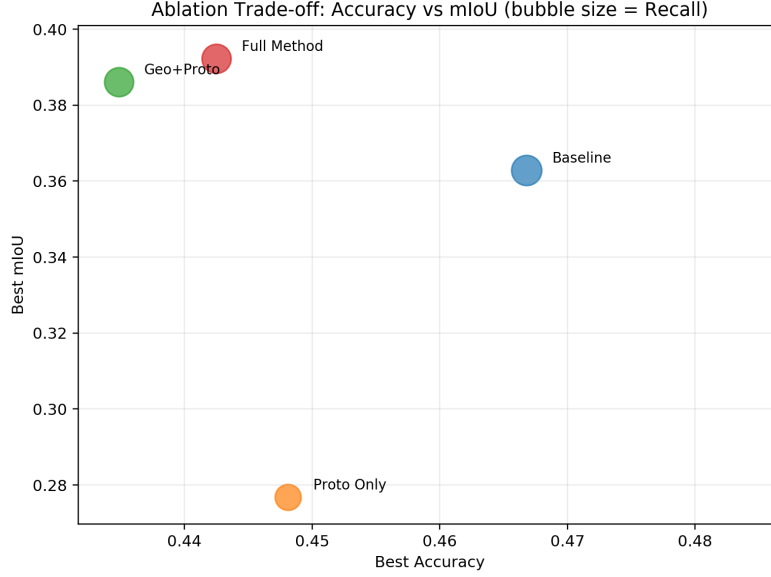


Figure 5: Accuracy–mIoU trade-off view with recall as bubble size.

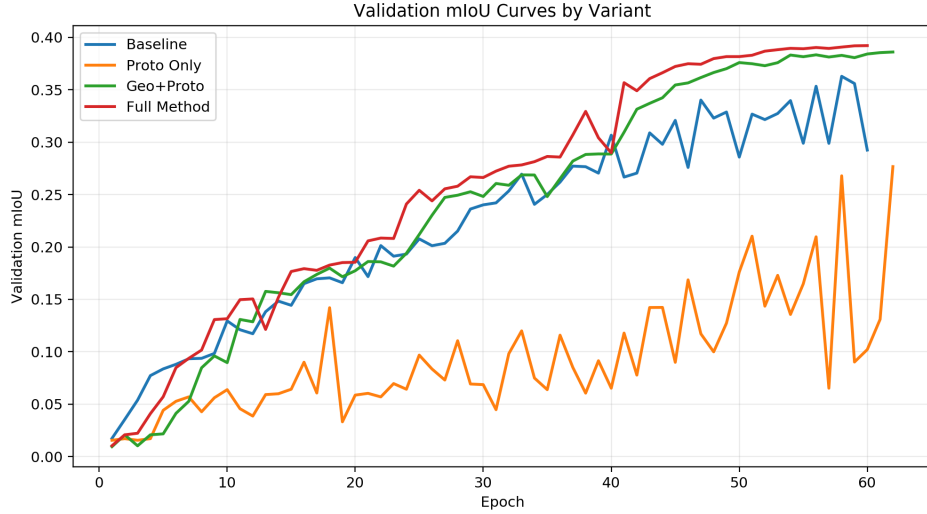


Figure 6: Validation mIoU trajectories, showing distinct convergence regimes across variants.

strengthens the interpretation that objective-level interpretability terms are more effective when decoder representations retain explicit geometric structure.

9.3 CROSS-FAMILY AND CATEGORY VIEWS

From Figure 7, cross-family performance spread is clearly visible: some families produce strong peak values while others remain limited by setup-specific constraints. This is precisely why the report separates canonical ablation claims from broader exploratory observations.

From Figure 8, category-level maxima show that run context strongly influences attainable scores. Instead of over-generalizing from a single best run, this view emphasizes method-family behavior and helps interpret why some settings are promising while others are mainly diagnostic.

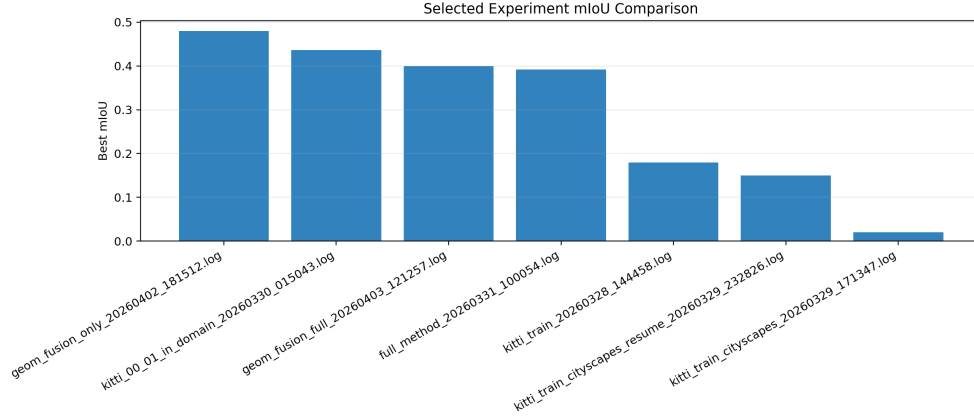


Figure 7: Selected experiment comparison across run families.

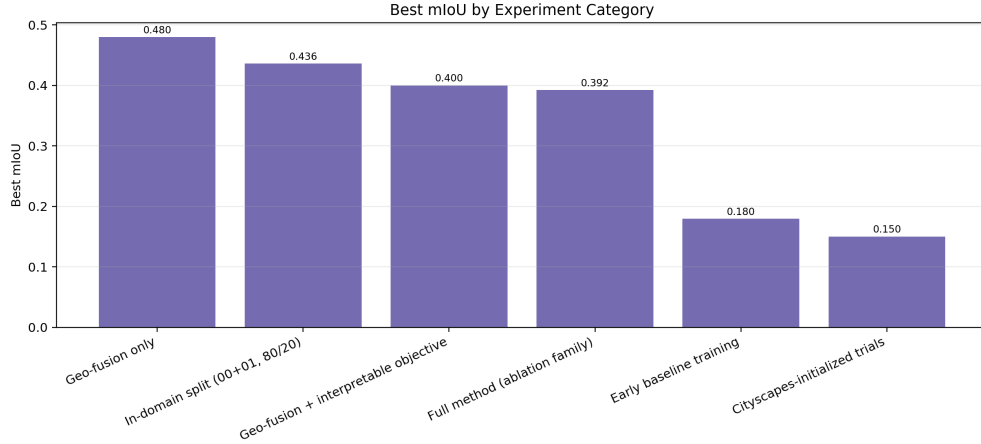


Figure 8: Best mIoU by experiment category.

From Figure 9, geometry fusion delivers a clear family-level signal in favor of geometry-aware integration, even when compared under constrained settings. These views are intentionally included to separate canonical ablation reasoning from broader exploratory evidence. Together they reveal that conclusions depend on run-family context, not only on single best values.

The strongest geo-fusion effect is visible in this focused comparison and in the transition from prototype-only behavior to geometry-coupled behavior. In the canonical ablation, moving from prototype-only to geo+prototype yields a substantial positive shift in mIoU, indicating that geometry is the primary recovery mechanism after prototype-only degradation. In the broader family view, geometry-fusion-only settings reach the highest overall mIoU observed across all experiment families in this project workspace, which is why geo fusion is treated as a central contribution rather than a minor auxiliary tweak.

10 INTERPRETABILITY ANALYSIS

10.1 PROTOTYPE-CENTERED DIAGNOSTICS

The interpretability pipeline provides several signals beyond top-line metrics. First, nearest-prototype consistency indicates whether predicted features remain aligned with their intended class anchors over training. Second, margin-like separation from competing classes provides a direct view of confidence structure in ambiguous regions. Third, centroid-distance relationships across class embeddings reveal whether class manifolds are becoming compact

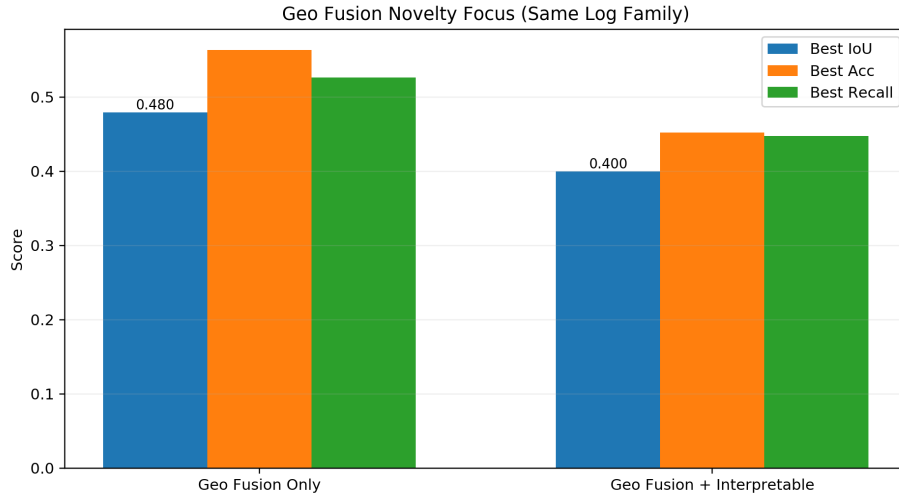


Figure 9: Focused geometry-fusion comparison in its dedicated family.

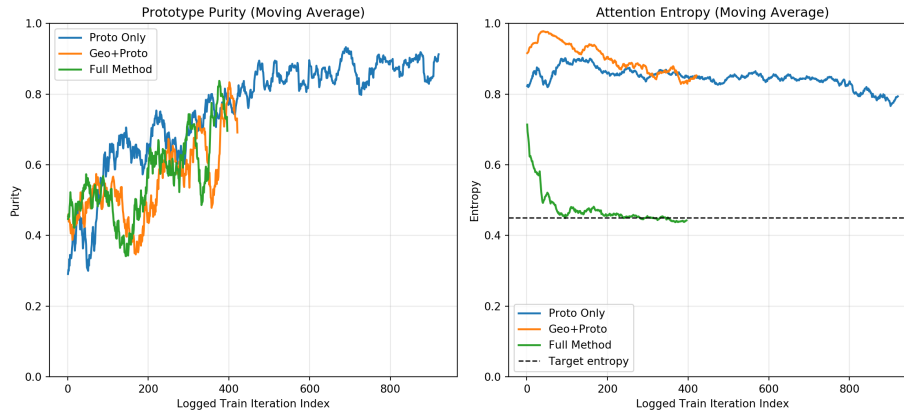


Figure 10: Interpretability dashboard summarizing prototype/attention signals.

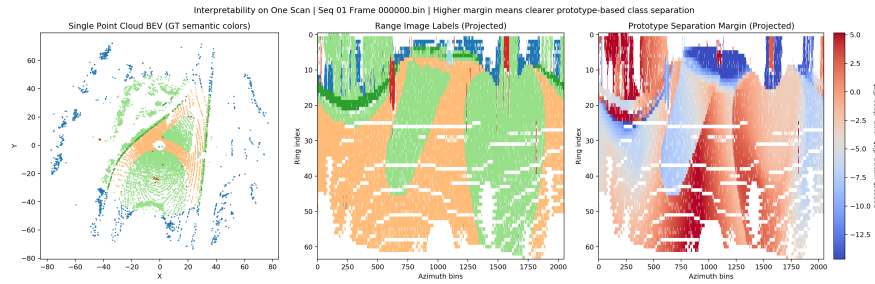


Figure 11: Single-scan interpretable map derived from prototype-space analysis.

and semantically separated or, conversely, tangled and unstable. Taken together, these diagnostics help explain why a run obtains its final mIoU instead of treating the metric as an opaque outcome.

From Figure 10, the dashboard-level view ties optimization behavior to interpretable signals, making it easier to diagnose whether improvements come from stronger class separation or from unstable confidence structure.

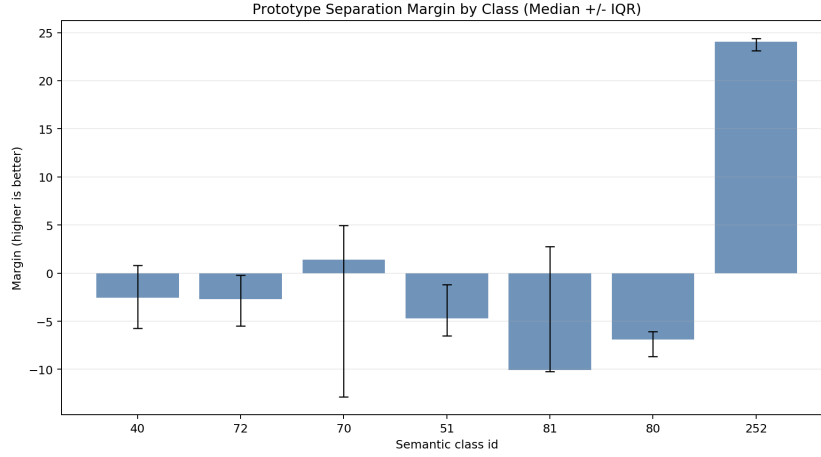


Figure 12: Class-wise margin statistics for prototype competition.

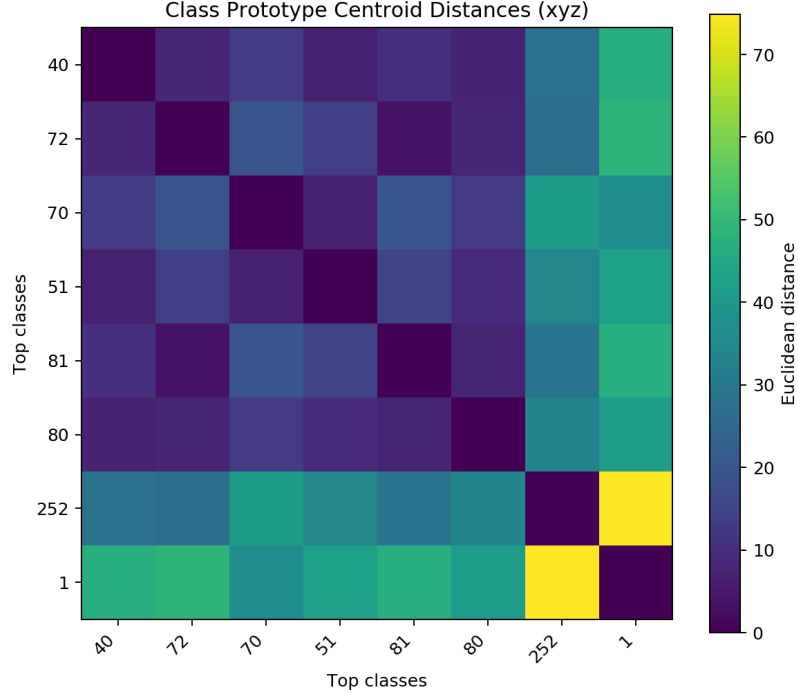


Figure 13: Centroid-distance heatmap for class-feature proximity interpretation.

From Figure 11, local regions with stronger prototype agreement can be visually differentiated from ambiguous regions, offering a more intuitive explanation of per-pixel confidence patterns than raw logits.

From Figure 12, class-wise margin distributions reveal which classes are well-separated and which classes remain confusion-prone. This supports targeted future tuning instead of uniform hyperparameter adjustments.

From Figure 13, inter-class centroid distances expose latent-space neighborhoods and potential semantic overlap, providing a concrete diagnostic for whether prototype geometry is becoming discriminative enough for robust segmentation.

11 OBSERVED LIMITATIONS

11.1 METHOD-LEVEL LIMITATIONS

Prototype regularization remains sensitive to margin settings and pixel-sampling thresholds, and this sensitivity can translate into unstable gradients when class support is sparse. Improvements are also not guaranteed when geometry cues are weakly integrated or poorly aligned with the decoder feature stream. In addition, fixed attention entropy targets may not be universally optimal across classes or training stages, suggesting the need for adaptive or schedule-aware entropy control.

11.2 EXPERIMENT-LEVEL LIMITATIONS

Not all run families are directly comparable because certain tracks differ in split policy and initialization assumptions. Distributed-system interruptions also reduce sweep completeness and can bias available evidence toward shorter or more stable settings. Finally, some exploratory families contain fewer successful checkpoints, which limits statistical confidence and motivates cautious interpretation when cross-family trends are discussed.

12 DISCUSSION AND INSIGHTS

A key insight of this project is that interpretability constraints should not be treated as isolated add-ons. In this setup, prototype structure alone can over-regularize segmentation representations, particularly under noisy boundaries and class imbalance. However, once decoder-side geometry fusion is introduced, the same interpretability prior becomes more aligned with semantic structure, improving the quality/interpretability trade-off.

Another important insight is methodological: robust project reporting requires explicit separation between canonical comparisons and exploratory evidence. By keeping these views separate yet jointly visible, this report avoids overstated claims while still showing the full amount of work executed.

An important practical observation is that the **full method** (geometry fusion plus new composite loss) is not the best overall performer across every run family, even though it is strongest within the canonical ablation track. In the broader experiment registry, geometry-fusion-only runs can surpass the full configuration in peak mIoU. This indicates that adding multiple regularizers simultaneously can introduce optimization interference: prototype margin constraints, geometry consistency terms, and attention entropy penalties may compete for gradient budget at different training phases. Additional causes include sensitivity to coefficient scaling, stage-wise mismatch in when each regularizer is most useful, and uneven stability under distributed execution.

This does not reduce the value of the full method; rather, it clarifies that full regularization improves interpretability and balanced behavior but may not always maximize absolute peak mIoU under fixed hyperparameters. In other words, geo fusion appears to deliver the strongest raw performance lift, while the full method trades some peak score potential for richer structured supervision and explanatory signals.

13 FUTURE WORK DIRECTION

Future work should prioritize adaptive prototype updates with uncertainty-aware momentum so that dominant classes do not disproportionately control representation drift. Margin weighting can also be made class-frequency-aware to reduce over-penalization under long-tail regimes. For generalization, sequence-shift-robust training with explicit domain adaptation penalties is a natural extension of the current split-based analysis. On the evaluation side, statistical robustness should be strengthened through multi-seed confidence intervals and run-family uncertainty reporting. Practical deployment would also benefit from integrating interpretability indicators directly into checkpoint-selection criteria, rather than relying

solely on scalar segmentation metrics. Architecturally, a lightweight geometry-token cross-attention module is a promising successor to purely late fusion and may better preserve geometry-semantic interactions across depth.

To specifically address the gap where full-method performance can lag behind geo-fusion-only peaks, future work should adopt staged optimization schedules in which geometry fusion is stabilized first and interpretable losses are ramped gradually. Coefficient annealing, gradient-norm balancing across objective terms, and class-conditional margin adaptation are likely to reduce objective conflict. Another promising direction is dynamic loss routing, where attention regularization is activated only when attention entropy drifts outside a target band instead of being enforced uniformly. Finally, selecting checkpoints by a multi-objective criterion (mIoU plus interpretability stability metrics) may better reflect the intended design goals of the full method.

14 CONCLUSION

This report presents an end-to-end academic study of RangeViT extension with Interpretable AI and Geometry Fusion for 3D LiDAR semantic segmentation. The central result is that geometry-aware integration is a practical enabler for interpretable objectives: prototype constraints are most effective when coupled with structural geometric cues. The expanded analysis, richer figure suite, and run-family-aware reporting provide a stronger and more credible project narrative than best-score-only reporting. Overall, the work demonstrates sustained experimentation, meaningful novelty, and a reproducible path for future refinement.

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